

Medical Evidence Graph & Outcomes Insight Lab

This document provides a detailed, implementation-focused description of the Medical Evidence Graph & Outcomes Insight Lab. The Lab is a Motia-orchestrated platform that consolidates medical evidence from literature, clinical trials, guidelines, registries, and de-identified outcomes data into an explainable knowledge graph and analytics environment. Its purpose is to enable clinicians, analysts, and decision-makers to answer 'what works for whom, under which conditions' with transparent provenance.

1. Problem & Vision

Healthcare decisions often rely on fragmented evidence: randomized trials on narrow populations, observational studies with diverse confounders, scattered guidelines, and siloed real-world data. Key challenges include:

- Difficulty synthesizing evidence across trials, guidelines, and real-world outcomes.
- Limited visibility into which interventions work best for specific sub-populations.
- Slow, manual workflows for building cohorts and performing comparative analyses.
- Lack of transparent, machine-readable representations of guidelines and care pathways.

The Medical Evidence Graph & Outcomes Insight Lab addresses these gaps by:

- Building a unified evidence graph linking conditions, interventions, populations, outcomes, and sources.
- Providing graph-aware retrieval so queries navigate relationships, not just keywords.
- Enabling cohort-centric outcomes analysis and comparative effectiveness at scale.
- Offering explainable outputs where every recommendation is backed by visible evidence paths.

2. Core Outcomes

The Lab is designed to achieve the following outcomes for participating organizations:

1) Evidence Consolidation

- Single, queryable fabric for trials, guidelines, publications, and curated real-world data.
- Harmonized vocabularies (ICD, SNOMED CT, LOINC, RxNorm) to align heterogeneous sources.

2) Cohort-Centric Insights

- Rapid definition of patient cohorts and observation windows.
- Standardized metrics for survival, readmission, complications, and cost.

3) Comparative Effectiveness & Safety

- Side-by-side evaluation of interventions, including subgroup analyses.
- Transparent indication of evidence strength and limitations.

4) Guideline & Pathway Support

- Machine-readable guidelines and care pathways linked to actual outcomes.
- Detection of deviations and opportunities for optimization.

5) Governance-Ready Analytics

- Built-in privacy protections, provenance tracking, and role-based access controls.

3. Architecture Overview

The Lab uses Motia as the workflow and event backbone, connecting ingestion, graph construction, retrieval, and analytics services into a coherent system.

Core components:

- Data Storage:
 - PostgreSQL for metadata, cohort definitions, and process logs.
 - Object storage (S3/MinIO) for raw artifacts: PDFs, XML, JSON, CSV.
- Search & Embeddings:
 - OpenSearch for structured and unstructured keyword search and filters.
 - Qdrant for dense vector search across abstracts, sections, and guideline fragments.
- Knowledge Graph Layer:
 - Neo4j capturing entities such as conditions, interventions, drugs, procedures, outcomes, populations, providers, and organizations, and the relationships among them.
 - PyKEEN / GNNs for link prediction (e.g., suggesting related interventions or populations).
- Analytics & Modeling:
 - PyTorch and survival analysis libraries for risk and outcome models.
 - Optional causal inference components (e.g., uplift models) where data quality permits.
- Security & Governance:
 - Vault for key and secret management.
 - OPA policies to constrain who can query which data sets and at what granularity.
 - Presidio for PHI detection and masking when ingesting semi-structured sources.
- Observability:
 - MLflow for experiment and model tracking.
 - Langfuse for tracing any LLM/RAG-based query flows.
 - Prometheus + Grafana + OpenSearch for metrics, logs, and traces.

4. Key Workflows

4.1 Evidence Ingestion & Normalization

- Motia flows fetch data from literature APIs, clinical trial registries, guideline repositories, and de-identified registries or claims extracts.
- Parsers extract structured fields (conditions, interventions, endpoints, populations) and map them to standard ontologies.
- Normalized records and their provenance are stored in PostgreSQL and object storage.

4.2 Evidence Graph Construction

- Entities are instantiated as nodes; relations as edges (e.g., Condition TREATS_WITH Intervention, Intervention HAS_OUTCOME Outcome for Population).
- KGE / GNN-based enrichment suggests additional plausible links and flags inconsistencies.
- Graph updates are triggered via Motia events when new evidence is ingested.

4.3 Graph-RAG Evidence Query

- Incoming clinical or analytic questions (via API/UI) are decomposed into:
 - Text search over documents (OpenSearch),
 - Embedding search (Qdrant),
 - Targeted graph traversals in Neo4j.
- A fusion layer merges and reranks results and returns:
 - Key passages,
 - Structured evidence paths in the graph,

- Source citations and confidence indicators.

4.4 Cohort & Outcomes Analytics

- Users define cohorts using clinical codes, demographics, time windows, and utilization rules.
- The system computes survival curves, event rates, and comparative outcomes across treatment arms.
- Where appropriate, statistical adjustments and sensitivity analyses are applied, and results are versioned for governance.

4.5 Guideline & Pathway Intelligence

- Guidelines are modeled as decision graphs; adherence is assessed against observed care.
- The platform highlights patterns where deviations correlate with better or worse outcomes, enabling iterative refinement of pathways.

5. Security, Privacy & Governance

Given the sensitivity of healthcare data, the Lab enforces:

- Use of de-identified or synthetic data for research by default unless governed agreements allow otherwise.
- PHI detection and redaction via Presidio for free-text inputs and imported documents.
- Strong authentication integrated with enterprise identity providers.
- Fine-grained authorization using OPA to restrict access by role, dataset, geography, and project.
- Full lineage: every dataset, transformation, model, and query is logged for audit.
- Deployment flexibility: single-tenant SaaS, dedicated VPC, or full on-prem installation.

6. Implementation & Extensibility

Phase 1: Stand up ingestion for literature, trials, and guidelines; basic evidence graph; basic search.

Phase 2: Add de-identified outcome datasets; cohort builder; survival and comparative analytics.

Phase 3: Full Graph-RAG experience with rich evidence paths and guideline/pathway modeling.

Phase 4: Advanced causal and subgroup analytics; integrations with partner EMR/registry systems under strict governance.

The architecture is modular: organizations can adopt only the evidence graph and search layer, or extend into full outcomes analytics and guideline optimization. The guiding principle is to deliver highly explainable, regulation-aligned analytics on top of a robust medical evidence graph.